



Chitosan nanoemulsion incorporated with *Carum carvi* essential oil as ecofriendly alternative for mitigation of aflatoxin B₁ contamination in stored herbal raw materials

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ABSTRACT

The present investigation entails the first report on entrapment of *Carum carvi* essential oil (CCEO) into chitosan polymer matrix for protection of stored herbal raw materials against fungal inhabitation and aflatoxin B₁ (AFB₁) production. Physico-chemical characterization of nanoencapsulated CCEO was performed through Fourier transform infrared spectroscopy, dynamic light scattering, X-ray diffractometry, and scanning electron microscopy. The nanoencapsulated CCEO displayed improved antifungal and AFB₁ suppressing potentiality along with controlled delivery over unencapsulated CCEO. The encapsulated CCEO nanoemulsion obstructed the ergosterol production and escalated the efflux of cellular ions, thereby suggesting plasma membrane as prime target of antifungal action in *Aspergillus flavus* cells. The impairment in methylglyoxal production and modeling based carvone interaction with Afl-R protein validated the antiaflatoxigenic mechanism of action. In addition, CCEO displayed augmentation in antioxidant potentiality after encapsulation into chitosan nanomatrix. Moreover, the *in-situ* study demonstrated the effective protection of *Withania somnifera* root samples (model herbal raw material) against fungal infestation and AFB₁ contamination along with prevention of lipid peroxidation. The acceptable organoleptic qualities of *W. somnifera* root samples and favorable safety profile in mice (animal model) strengthen the application of nanoencapsulated CCEO emulsion as nano-fungitoxicant for preservation of herbal raw materials against fungi and AFB₁ mediated biodeterioration.

1. Introduction

Herbal raw materials are an integral part of traditional medicinal system particularly in European and Asian countries with quite popularity among people. The global demands of traditional herbal based medicines are gaining momentum due to potent anti-inflammatory, anti-convulsant, nephroprotective, anti-diabetic, anti-spasmodic, and cardioprotective properties, and can be regarded as a proxy to the allopathic medicines with lots of side effects and microbial resistance [1]. Among different herbal raw materials, *Withania somnifera* roots, commonly known as Indian ginseng, are of great pharmacological significance against neurodegenerative disease, cancer, stress, and

cardiovascular problems. Most notably, the roots are of multidimensional significance due to the presence of withanolides (steroid lactone), withaferone, and withanone showing immunomodulatory, aphrodisiac, and antiviral properties [2]. However, herbal raw materials are prone to fungal contamination during postharvest storage, processing, and transportation [3]. *Aspergillus flavus* is a common filamentous fungus causing contamination of herbal raw materials, especially *W. somnifera* roots by production of aflatoxin B₁ (AFB₁) in congenial environmental conditions of subtropical belts [4]. Most remarkably, the fungal and AFB₁ contamination adversely affect the active principles of herbal drugs and diminish their characteristic therapeutic potency [5–7]. Most importantly, AFB₁ has gained the maximum attention to research

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community due to its carcinogenic, hepatotoxic, mutagenic, and immuno-suppressive effects on both human and animals and is designated as Group-1 human carcinogen by International Agency for Research on Cancer (IARC) [8]. Hence, there is high-priority to propose some novel and safer strategies to protect the stored herbal raw materials from mycotoxins, especially AFB₁ contamination.

Varieties of synthetic preservatives are available to combat the fungal and AFB₁ contamination in stored herbal raw materials, however, their global appreciation are restricted due to residual toxicity to non-target organism, eco-toxicity affecting food chain, environmental persistence, and mammalian toxicity [9–11]. Currently, the growing interest in industries is to utilize some novel and non-synthetic alternatives for improving the shelf-life of stored herbal drugs, and preserve their active principles. In this reference, plant based essential oils (EOs) exhibit maximum potentiality to combat fungal and AFB₁ spoilage in stored herbal system. Moreover, EOs are grouped under “Generally Recognized as Safe” (GRAS) category, demonstrating safe use over synthetic antimicrobials [12]. In spite of several advantages, applications of EOs are limited due to rapid volatility, less stability, hydrophobicity, and intense aroma causing alteration of sensory profile of herbal raw materials [13,14]. In order to overcome these obstacles, nanoencapsulation of EOs into carrier matrices (polysaccharides, proteins, and lipids) offers an innovative strategy to maintain the stability, reduce volatility, improve the biological efficacy, and protect from external environmental conditions [15]. Among different polymeric matrices, chitosan (polysaccharide) has strong evidences of EOs entrapment due to its non-toxic, biodegradable, biocompatible nature, and also inclusion under GRAS category [16]. Among different encapsulation techniques employed to entrap EOs, ionic gelation based nanoemulsion fabrication has been recognized as the most effective technique. The ionic gelation process is mainly utilized in pharmaceutical and drug delivery due to less utilization of chemicals, and development of particles within the range of 100 nm [17].

Carum carvi L. (Family: Apiaceae) commonly called as ‘caraway’ is an aromatic annual herbaceous plant, native to the central Asia, Northern and Central Europe. The essential oil (CCEO) obtained from dried fruits showed anti-diarrheal, antispasmodic, anti-flatulence, and anti-dysmenorrheal activities [18]. Few reports on antifungal activity of CCEO are available; however, no research has been demonstrated pertaining to nanoencapsulation of CCEO and its practical application for protection of herbal raw material against aflatoxigenic *A. flavus*, AFB₁ secretion, and preservation of active principles with emphasis on possible mechanism of action. On the basis of this background, the objective of the study was encapsulation of CCEO within chitosan nanoemulsion, physico-chemical characterizations, and application of CCEO nanoemulsion as controlled release delivery system for protection of *W. somnifera* root samples against *A. flavus* inhabitation, and AFB₁ production. The present work has elucidated the target site of fungal inhibition and AFB₁ suppressing mechanism of action. Additionally, *in situ* protection activity against AFB₁ mediated lipid peroxidation in *W. somnifera* roots, and safety evaluation in mice has been conducted in order to provide a green insight for application of encapsulated CCEO nanoemulsion as green fungitoxicants in pharmaceutical and agricultural industries.

2. Materials and methods

2.1. Chemicals

Low molecular weight chitosan (Cs) (deacetylation degree >85%) dimethyl sulfoxide (DMSO), perchloric acid (HClO₄), silica gel-G, Tween 80, Tween 20, sodium tri polyphosphate (S-TTP), 1,2-diaminobenzene, dichloromethane (DCM), methanol, glacial acetic acid, methylglyoxal (MG), acetonitrile, trichloroacetic acid (TCA), K₂CO₃, hydrochloric acid (HCl), thiobarbituric acid (TBA), DPPH (2,2-diphenyl-1-picrylhydrazyl), and ABTS (2,2'-Azino-bis [3-ethylbenzothiazoline-6-

sulfonic acid] diammonium salt) (purity grade >98%) were purchased from Sigma labs, Maharashtra, India. Culture media viz. DPA (dextrose 20 g, potato 200 g, agar 15 g) and YMSK (Yeast extract 7 g; MgSO₄ 0.5 g; Sucrose 200 g; KNO₃ 0.3 g) were supplied by Sisco Research Laboratories Maharashtra, India.

2.2. Fungal species

Fungal species viz. *Aspergillus repens*, *A. niger*, *A. luchuensis*, *A. sydowii*, and *Penicillium purpurogenum* along with AFB₁ producing strain of *Aspergillus flavus* was used in the current investigation [11,19]. Spore suspension of different fungal strain was prepared from seven days old culture and spores were diluted in Tween 80 solution (0.1 %). The spore density (10⁶ spores/mL) was maintained by haemocytometer.

2.3. Extraction and characterization of *Carum carvi* essential oil (CCEO)

For extraction of CCEO, 1 kg dry seeds of *C. carvi* was introduced to Clevenger's apparatus and suspended into 3 L of water followed by heating at 70 °C for 1 h. Thereafter, the temperature of the Clevenger's heating mantle was set at 50 °C over a period of 3 h. The essential oil components were distilled with water vapor, condensed to liquid state and subsequently separated from the top of the condensed water. Thereafter, the collected essential oil was dried over anhydrous sodium sulfate to remove surface water. The CCEO was isolated in amber vial and kept in refrigerator at 4 °C. Chemical profile of CCEO was investigated by Gas chromatography (GC) fitted with mass spectrometry (MS). The column was Equity-5 (0.35 mm × 60 m × thickness 0.25 μm). The temperature of oven was laid down between 80 and 230 °C with 2 min of hold time. The detector and injector temperatures were 280 and 290 °C. Carrier gas was Helium at a flow rate of 1 mL/min. Source and transfer line temperature was set at 220 °C. The energy of ionization and mass scan range were developed at 70 eV and 40–400 amu. The CCEO characterization was attained with emphasis on elution order, retention time, and retention index employing n-alkane series (C₉–C₂₈) of hydrocarbons and National Institute of Standard and Technologies (NIST) 2011 library [20].

2.4. Preparation of CCEO incorporated chitosan nanoemulsion (CCEO-Ne)

Synthesis of CCEO-Ne was performed through two step protocol of oil-in-water emulsification and ionic-gelation suggested by Hadidi et al. [21] with moderate modification. For this, acetic acid dissolved chitosan solution (1 %) was developed and agitated for 12 h at room temperature. Thereafter, tween 80 (0.08 g) was mixed with chitosan solution and pH was maintained to 4.2 by stirring for 1 h at 40 °C to develop homogeneous solution. Requisite volume of DCM dissolved CCEO was added to chitosan solution at various ratios (chitosan/CCEO (w/v) 1:0, 1:0.2, 1:0.4, 1:0.6, 1:0.8 and 1:1) during homogenization at 12,560 ×g (under ice bath condition) for 25 min to develop oil-in-water emulsion. Subsequently, sodium tripolyphosphate (S-TTP) (0.40 %) was added to homogenized chitosan solution to promote ionic gelation and shaken for 40 min at 25–27 °C. Centrifugation of the developed particles was executed at 10,235 ×g for 10 min and resulting pellets were thoroughly washed. Finally, the pellet was liquefied with 10 mL of deionized water and sonicated (10 min) to break the aggregated particles. Chitosan nanoemulsion (Cs—Ne) was prepared in a similar way without mixing of CCEO.

2.5. Characterization of Cs—Ne and CCEO-Ne

The size of particles along with stability and polydispersity index (PDI) of Cs—Ne and CCEO-Ne was elucidated through dynamic light scattering (DLS) at 25 °C with 90° scattering angle and wavelength of 532 nm [22]. Morphological and structural characteristics of prepared

Cs—Ne and CCEO—Ne was examined by applying one drop of sonicated nanoemulsion on cover slip and further coated with gold. After that, the samples were observed in scanning electron microscope (SEM) in the scanning range 50,000 to 60,000 $\times$ [23]. Fourier transform infrared spectroscopy (FTIR) was done to analyze the functional groups of Cs, Cs—Ne, CCEO, and CCEO—Ne in between wave number 500 to 4000 cm^{-1} (4 cm^{-1} resolution) [24]. X ray diffraction (XRD) patterns of Cs, Cs—Ne, and CCEO—Ne were recorded between 2θ ranges 5° to 50° with speed of 5° per minute to study their crystallographic profile [23].

2.6. Elucidation of encapsulation and loading efficiency

The encapsulation (EE) and loading efficiency (LE) of CCEO—Ne was estimated by considering the amount of CCEO entrapped in chitosan using UV–Visible spectrophotometer [25]. In total, 300 μL of CCEO—Ne

$$\text{AFB}_1 (\mu\text{g/mL}) = \frac{\text{Absorbance at 360 nm} \times \text{Molecular mass (312) of AFB}_1}{\text{Molar extinction coefficient (21,800)} \times \text{Path Length (1 cm)}} \times 1000 \quad (4)$$

was homogenized in 2700 μL ethyl acetate (5 min) pursued by centrifugation at $12,790 \times g$ (11 min). Then, evaluation of the amount of CCEO loaded in Cs—Ne was carried out by extracting CCEO in ethyl acetate and compared with CCEO calibration curve constructed at 306 nm. Following formulae were used for calculation of EE and LE.

$$\text{EE (\%)} = \frac{\text{Amount of loaded CCEO in nanoemulsion}}{\text{Initial amount of CCEO added}} \times 100 \quad (1)$$

$$\text{LE (\%)} = \frac{\text{Amount of loaded CCEO in nanoemulsion}}{\text{Weight of nanoemulsion}} \times 100 \quad (2)$$

2.7. The release assay

The mechanism of CCEO release was performed in acetate (pH 3.0) and phosphate buffer saline (PBS, pH 7.0) considering the strategy of Das et al. [10] with moderate alteration. Briefly, 0.050 mL of CCEO—Ne was dissolved into 3.4 mL PBS mixed 100 % ethanol and acetate buffer and kept for 120 h. At specified time intervals, CCEO—Ne mixed buffer was centrifuged at $9870 \times g$ (11 min) and 0.020 mL of upper layer was withdrawn for the analysis. The remaining volume of reaction solution was fulfilled by PBS and acetate buffer. CCEO release in PBS and acetate buffer medium was calculated at 306 nm by the following equation.

$$\text{Cumulative release (\%)} = \frac{\text{Cumulative mass of released CCEO at each time}}{\text{Initial weight of CCEO loaded in the sample}} \times 100 \quad (3)$$

2.8. Antifungal and anti-AFB₁ effectiveness of CCEO and CCEO—Ne: In vitro evaluation

In vitro antifungal assessment of CCEO and CCEO—Ne was investigated against the *A. flavus*, *A. repens*, *A. niger*, *A. sydowii*, *A. luchuensis*, and *Penicillium purpurogenum* using broth dilution technique of Singh et al. [26]. For this, 20 mL of YMKS medium was homogenized with requisite amount of CCEO and CCEO—Ne in conical flask followed by inoculation of 0.001 mL spore suspension (10^6 spore/mL adjusted by

haemocytometer) of different fungal strain and kept for incubation at $27 \pm 2^\circ\text{C}$ for 10 days. In control set, CCEO and CCEO—Ne were not added. After 10 days of incubation, the lowest dose of CCEO and CCEO—Ne bringing 100 % inhibition of fungal proliferation were considered as the minimum inhibitory concentrations (MICs).

To assess the anti-AFB₁ potentiality of CCEO and CCEO—Ne, the medium was aseptically mixed with chloroform (20 mL) and gently agitated for 2 min. The lower chloroform layer was separated and dried at 95°C . The left material was homogenized into methanol and 0.005 mL solution was spotted on thin layer chromatography (TLC) plate using the mobile phase of isoamyl alcohol, methanol, and toluene solution. The blue spots of AFB₁ was scrapped and added to methanol (5 mL) followed by centrifugation at $4872 \times g$ for 5 min. AFB₁ absorbance was noted at 360 nm and the amount was calculated by the following equation.

Concentration representing the 100 % percent inhibition of AFB₁ synthesis was marked as minimum AFB₁ inhibitory concentration (MAIC).

2.9. Antifungal mechanism of action of CCEO—Ne against *A. flavus*

The mechanism of action underlying the fungitoxic activity of CCEO—Ne against *A. flavus* cells was investigated in terms of ergosterol reduction, release of cell contents, and membrane integrity. The effect of CCEO—Ne on ergosterol synthesis in cell membrane was quantified using the strategy of Tian et al. [27]. Briefly, 0.003 mL spore suspension of *A. flavus* (10^6 spore/mL) was dispersed into 20 mL YMKS medium in conical flask with requisite concentrations of CCEO—Ne (0.05, 0.10, 0.15, 0.20, and 0.25 g/L) and cultured for 4 days ($27 \pm 2^\circ\text{C}$). As a control, YMKS media devoid of CCEO—Ne was used. After that, the grown biomass was transferred to 5 mL ethanol-KOH solution, agitated for 3 min, and subjected to heating at $70 \pm 2^\circ\text{C}$ for 40 min. Ergosterol was extracted from mature biomass by addition of aqueous n-heptane solution and agitated for 3 min. Subsequently, the upper fresh upper layer was collected and used for ergosterol assessment by scanning spectrophotometry between 230 and 300 nm using the following formula [28].

$$\% \text{Ergosterol} + \%24(28) \text{ dehydroergosterol} = \frac{\left(\frac{\text{Absorbance } 282 \text{ nm}}{290} \right)}{\text{Mycelial mass}} \quad (5)$$

$$\%24(28) \text{ dehydroergosterol} = \frac{\left(\frac{\text{Absorbance } 230 \text{ nm}}{518} \right)}{\text{Mycelial mass}} \quad (6)$$

$$\% \text{Ergosterol} = \% \text{Ergosterol} + \%24(28) \text{ dehydroergosterol} - \%24(28) \text{ dehydroergosterol} \quad (7)$$

The strategy of Helal et al. [29] and Jenie et al. [30] was employed for evaluating vital cellular constituent's leakage in terms of ions (Mg^{2+} , K^+ , and Ca^{2+}) and 260 nm (nucleic acids), 280 nm (protein) absorbing materials. For this, 7 days old culture of *A. flavus* was treated with requisite doses (0.05, 0.10, 0.15, 0.20, and 0.25 g/L) of CCEO-Ne for 16 h. Thereafter, extracellular ions were analyzed by atomic absorption spectroscopy and determination of nucleic acids and protein leakage was performed using the optical density reading of supernatant through UV-visible spectrophotometry.

The membrane integrity was performed by using the propidium iodide (PI) binding assay. For this 0.3 g of *A. flavus* biomass was treated with 0.05, 0.10, 0.15, 0.20, and 0.25 g/L concentrations of CCEO-Ne and kept in B.O.D incubator for 16 h. The biomass was suspended to PBS and stained with PI dye for 30 min [31]. Subsequently, the biomass was washed and crushed in PBS (2 mL) followed by centrifugation at $5200 \times g$ for 12 min. The upper transparent layer was used for recording of absorption by fluorescent spectroscopy at excitation and emission wavelength of 546 and 590 nm, respectively.

2.10. The biochemical and molecular mechanism of AFB₁ suppression

2.10.1. Methylglyoxal assessment

AFB₁ suppressing action mechanism of CCEO-Ne was carried out in terms of methylglyoxal assessment in *A. flavus* cells. In total, 0.3 g of seven days old fungal mat was treated with requisite doses (0.05, 0.10, 0.15, 0.20, and 0.25 g/L) of CCEO-Ne in YMKS medium and incubated for 12 h at $25 \pm 2^\circ C$. Control set did not contain CCEO-Ne in YMKS medium. Thereafter, the fungal mycelia were crushed into 500 mM $HClO_4$, kept at ice box ($4^\circ C$, 13 min) and centrifuged at $13,621 \times g$ for 13 min. The pH of upper layer was maintained to 7.0 by addition of K_2CO_3 solution followed by centrifugation at $11,208 \times g$ for 10 min. For cellular methylglyoxal determination, 0.65 mL supernatant was added to 0.25 mL of 1,2 diaminobenzene, and 0.1 mL of 5 M perchloric acid followed by measuring the absorbance at 341 nm using UV-Visible spectrophotometer [32].

2.10.2. In silico interaction study

$$\text{Protection potentiality against total fungi} = \frac{\text{Fungal colony number in control root samples} - \text{Fungal colony number in treated root samples}}{\text{Fungal colony number in control root samples}} \times 100 \quad (9)$$

$$\text{Protection potentiality against } A. \text{flavus} = \frac{A. \text{flavus colony number in control root samples} - A. \text{flavus colony number in treated root samples}}{A. \text{flavus colony number in control root samples}} \times 100 \quad (10)$$

Amino acid sequence (FASTA format) of Afl-R was collected from NCBI site and three dimensional (3 D) structure of Afl-R was developed by Phyre 2 software. The 3 D structure of carvone was obtained from PubChem server. At the time of modeling, all parameters were set as default and structure of protein was validated by PDBSum. For molecular docking study, a combination of Swiss dock, Chimera and Pymol software were required. Ligand (carvone) and receptor protein (Afl-R) interaction was confirmed by PathDock web server. The results of PatchDock server was refined by FireDock server to get 10 best models on the basis of interactive energy like global energy, attractive vander waal's force, repulsive vander waal's force, and atomic contact energy (ACE). The interaction of Ligand and receptor protein was determined by Discovery Studio 4.0 and Chimera 1.8.1 [33].

2.11. Antioxidant activity of CCEO-Ne

Free radical neutralizing capacity of CCEO and CCEO-Ne was evaluated by ABTS^{•+} and DPPH[•] assay with slightly modified protocol of Chaudhari et al. [34]. For ABTS^{•+} assay, varying concentrations of CCEO, Cs—Ne and of CCEO-Ne were suspended in ABTS^{•+} stock solution (1.9 mL) and mixture was kept apart from light for 6 min followed by absorbance determination at 734 nm through UV-visible spectrophotometry. For DPPH[•] assay, different concentrations of CCEO, Cs—Ne and CCEO-Ne were added to DPPH stock solution (0.004 %, 5 mL) and placed in dark condition for 20 min. The control set did not possess CCEO-Ne. The absorbance of the samples was measured at 517 nm through UV-Visible spectrophotometry. Percent scavenging of free radicals was determined in terms of IC₅₀ value (50 % radical scavenging capacity) by the following formula given by Chaudhari et al. [34].

$$IC (\%) = \frac{\text{Blank absorbance} - \text{Sample absorbance}}{\text{Blank absorbance}} \times 100 \quad (8)$$

here, IC = Inhibitory concentration.

2.12. In situ efficacy of CCEO-Ne in stored herbal raw material (*Withania somnifera* roots)

For assessing the *in situ* antifungal activity, 500 g of *Withania somnifera* roots (model herbal raw material) were stored in different storage containers (1 L). The uninoculated and inoculated controls were developed as per the introduction of 1 mL spore suspension of *A. flavus* in root samples. Controls were not treated with Cs—Ne and CCEO-Ne, while the treatment sets were fumigated with CCEO-Ne at 0.25 g/L concentration (MIC dose). The incubation of control and treatment sets was performed at $25 \pm 2^\circ C$ (RH ~ 75 %) for 1 year. After storage, the root samples (both treated and untreated) were crushed in sterile mixture grinder and 500 mg of milled sample was homogenized into 4.5 mL of deionized water. One milliliter of 10^{-3} dilution was introduced to 10 mL of DPA medium and left at $25 \pm 2^\circ C$ in B.O.D incubator for 10 days. Number of *A. flavus* colonies was recorded and protection potentiality was calculated by given formula.

To determine the AFB₁ content, 3 g of grinded root samples were properly mixed with methanolic water in conical flask and agitated for 1 h in mechanical shaker (500 ×g). Subsequently, samples were centrifuged at $10,230 \times g$ for 15 min and upper fresh layer was collected followed by mixing with 300 μL chloroform and 3 mL of KBr mixed water (6 %). Thereafter, the samples were centrifuged at $8600 \times g$ for 15 min and the lower portion was separated in Eppendorf's tube and dried over water bath in the steam of N₂ at $75^\circ C$. The left material was diluted in 1000 μL of HPLC grade methanol and injection of 5 μL of sample was done into C-18 column of high performance liquid chromatography with isocratic solution of methanol, acetonitrile, and water (17 mL + 19 mL + 64 mL). The flow rate was 1.2 mL/min. AFB₁ was detected at 360 nm [35]. AFB₁ standard curve in between 0.25 and 10 ng/μL was developed to quantify the amount of AFB₁ in *W. somnifera*

Table 1
Chemical characterization of *Carum carvi* essential oil through GC–MS analysis.

S. no.	Compounds	Retention time (min.)	Percent area
1	α -Pinene	6.83	1.13
2	β -Pinene	7.34	0.79
3	<i>p</i> -Cymene	7.86	3.12
4	β -Myrcene	8.2	0.13
5	Limonene	8.98	13.62
6	1,8-Cineole	9.58	0.63
7	α -Thuyone	10.32	0.22
8	β -Linalool	10.7	4.26
9	Cis dihydrocarvone	14.42	0.21
10	Trans carveol	14.91	0.2
11	Carvone	15.68	69.85
12	Perilla aldehyde	16.52	0.16
13	Octadecane	20.67	0.17
14	Phenol. 3.5-di-tert-butyl	20.93	0.11
15	Octadecane	21.32	0.11
16	Octadecane	23.43	0.11
17	Trans-Hex-2-enal	23.81	0.12
18	Undecane	24.13	0.11
	Total		95.05 %

Note: Components in the bold letters are major compounds.

roots by photo diode array detector. The protection of root samples against AFB₁ contamination was determined by the following formula.

$$\% \text{AFB}_1 \text{ protection} = \frac{\text{AFB}_1 \text{ content in control root samples} - \text{AFB}_1 \text{ content in treated root samples}}{\text{AFB}_1 \text{ content in control root samples}} \times 100 \quad (11)$$

2.13. Effect of CCEO-Ne on lipid peroxidation of *W. somnifera* roots

One gram of grinded *W. somnifera* root sample (treated and control) was dissolved in TBA-TCA-HCl reagent mixture (prepared by subsequent mixing of 15 % TCA, 0.375 % TBA and 0.25 N HCl) followed by evaporation on water bath at 95 °C. Then, cooling of the samples was done in running tap water and centrifuged at 5250 ×g for 20 min. Supernatant was transferred in Eppendorf's tube and absorbance was noted at 600 and 532 nm [36]. Extent of lipid peroxidation was measured in terms of malondialdehyde (MDA) content. The amount of MDA in root samples was expressed as μM/gFW on the basis of molar extinction coefficient of MDA (156 mM⁻¹ cm⁻¹).

2.14. Impact of CCEO-Ne on organoleptic qualities of *W. somnifera* roots

Organoleptic property of *W. somnifera* root samples (treated and control) was evaluated by five point hedonic scale test. Parameters of organoleptic properties were color, flavor, texture, and odor. The hedonic scale was arranged in the form of 1 = Most dislike, 2 = Moderate dislike, 3 = Neither like/dislike, 4 = Moderate like, and 5 = Most like [37]. Fifteen panelists from Banaras Hindu University, India in between the age group 25–50 years were involved for testing of organoleptic properties. A consent form was signed by panelists for evaluation of nanoemulsion fumigated *W. somnifera* root samples. Two minutes were given to each panelist for assessing the score and then asked to note on score card. Water was given in between the testing of two different samples to neutralize the taste.

2.15. Evaluation of LD₅₀ value of CCEO-Ne on mice (animal model)

The method of Dwivedy et al. [38] with little changes was used for

Table 2
Encapsulation efficiency and loading efficiency of CCEO-Ne.

Chitosan/CCEO (w/v)	Encapsulation efficiency (%)	Loading efficiency (%)
1:0	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a
1:0.2	56.37 ± 3.87 ^b	0.98 ± 0.015 ^b
1:0.4	70.63 ± 5.18 ^c	1.67 ± 0.11 ^c
1:0.6	84.26 ± 3.47 ^d	3.75 ± 0.54 ^d
1:0.8	95.52 ± 5.38 ^e	6.23 ± 1.52 ^e
1:1	83.64 ± 4.96 ^f	4.52 ± 1.38 ^f

Note: Values are mean (n = 3) ± standard error, the means followed by same letter in the same column are not significantly different according to ANOVA and Tukey's multiple comparison test.

oral acute toxicity assay of CCEO-Ne. The toxicity evaluation was conducted in male mice having the approval certificate given by ethical and animal care committee of Banaras Hindu University. The mice were collected from Department of Zoology, Banaras Hindu University, India. The average weight of mice was 31.63 g (3 months old). Acclimatization of the mice was done for one week in the laboratory. Thereafter, required volumes of CCEO-Ne were subjected to oral administration to each mice group (10 mice) on the basis of catheter dependent micro-pipette. The control mice groups were not administered by CCEO-Ne. Observation of dead mice was performed within 48 h of CCEO-Ne administration. LD₅₀ (50 % lethality of mice) of CCEO-Ne was calcu-

lated on the basis of number of dead mice and the value was determined through probit analysis [39].

2.16. Statistical analysis

Each experiment was performed in a replicate of three and data sets were maintained as mean ± standard error along with one way analysis of variance (ANOVA) coupled to Tukey's B coefficient and significant differences at *P* < 0.05. Graphs were prepared in Microsoft Windows Excel 2019, Sigma Plot, and SPSS (Version 18.0, SPSS Inc. Chicago, USA) software.

3. Results and discussion

3.1. Extraction and characterization of CCEO

The CCEO extraction was performed through hydrodistillation of dried seeds and yield was recorded as 14 mL/kg of seeds. Chemical profiling of CCEO showed a total of 18 components comprising 95.05 % of total area. Among them, carvone (69.85 %), limonene (13.62 %), and β -linalool (4.26 %) were found as the major ingredients of CCEO (Table 1). Our result shows similarity to the view of Laribi et al. [40] demonstrating carvone as major component in German (77.35 %), Tunisian (76.37 %), and Egyptian varieties (61.38 %) of CCEO. However, variation in percent availability was mainly associated with genetic factors, geographical alteration, edaphic factors, and ecotype distribution [41]. It is worth to note that limonene was also accompanied by high proportion because it is an intermediate component in biosynthesis of carvone during secondary metabolism [42]. Therefore, chemical profiling of essential oil is a preliminary point before preparation of antifungal nanoemulsion and their application in commercial sectors.

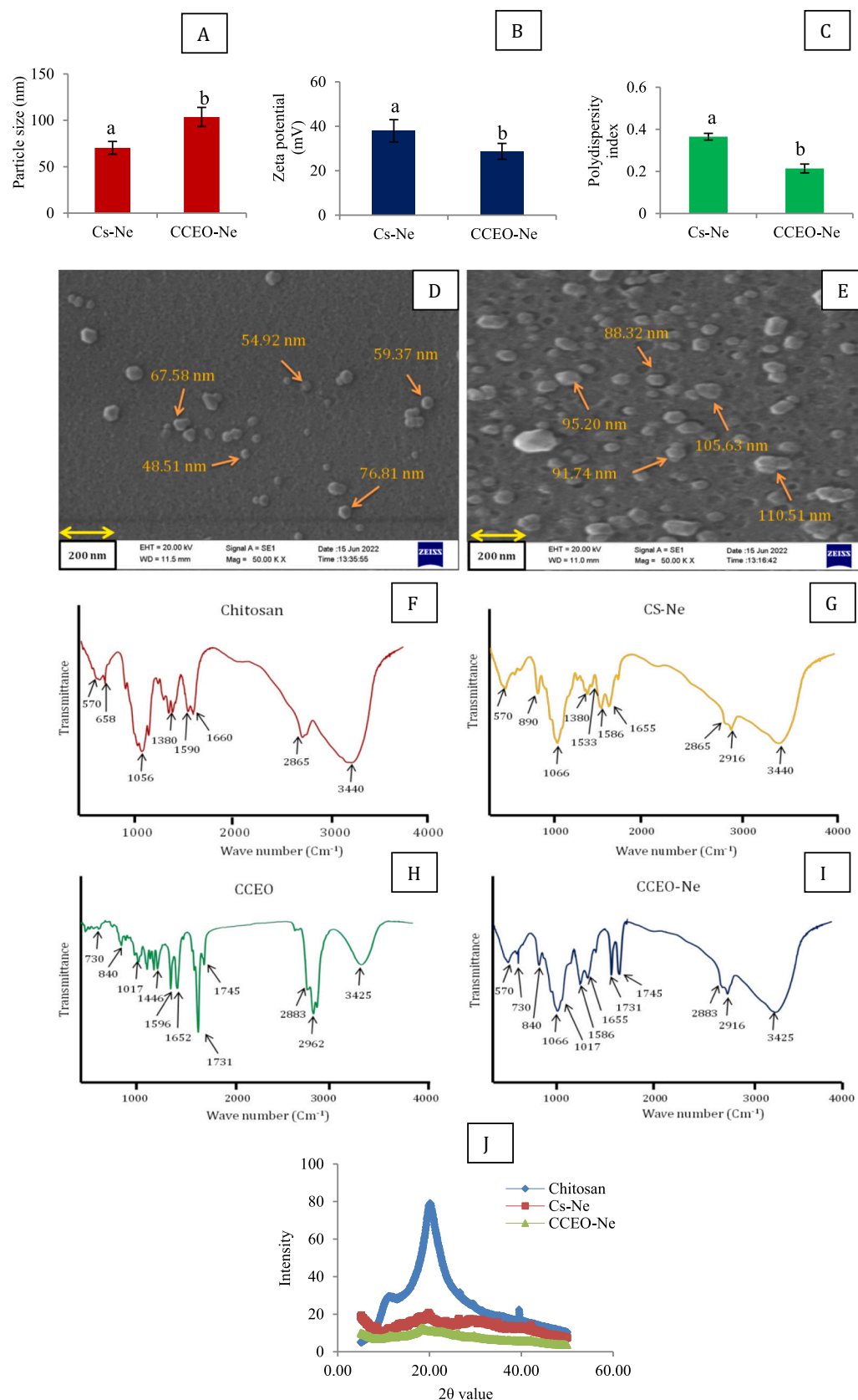


Fig. 1. (A) Particle size of Cs-Ne and CCEO-Ne, (B) zeta potential of Cs-Ne and CCEO-Ne, (C) polydispersity index of Cs-Ne and CCEO-Ne, (D) scanning electron microscopy (SEM) of Cs-Ne, (E) SEM of CCEO-Ne, (F, G, H, I) FTIR spectra of chitosan, Cs-Ne, CCEO, and CCEO-Ne, (J) X-ray diffractometry of chitosan, Cs-Ne, and CCEO-Ne.

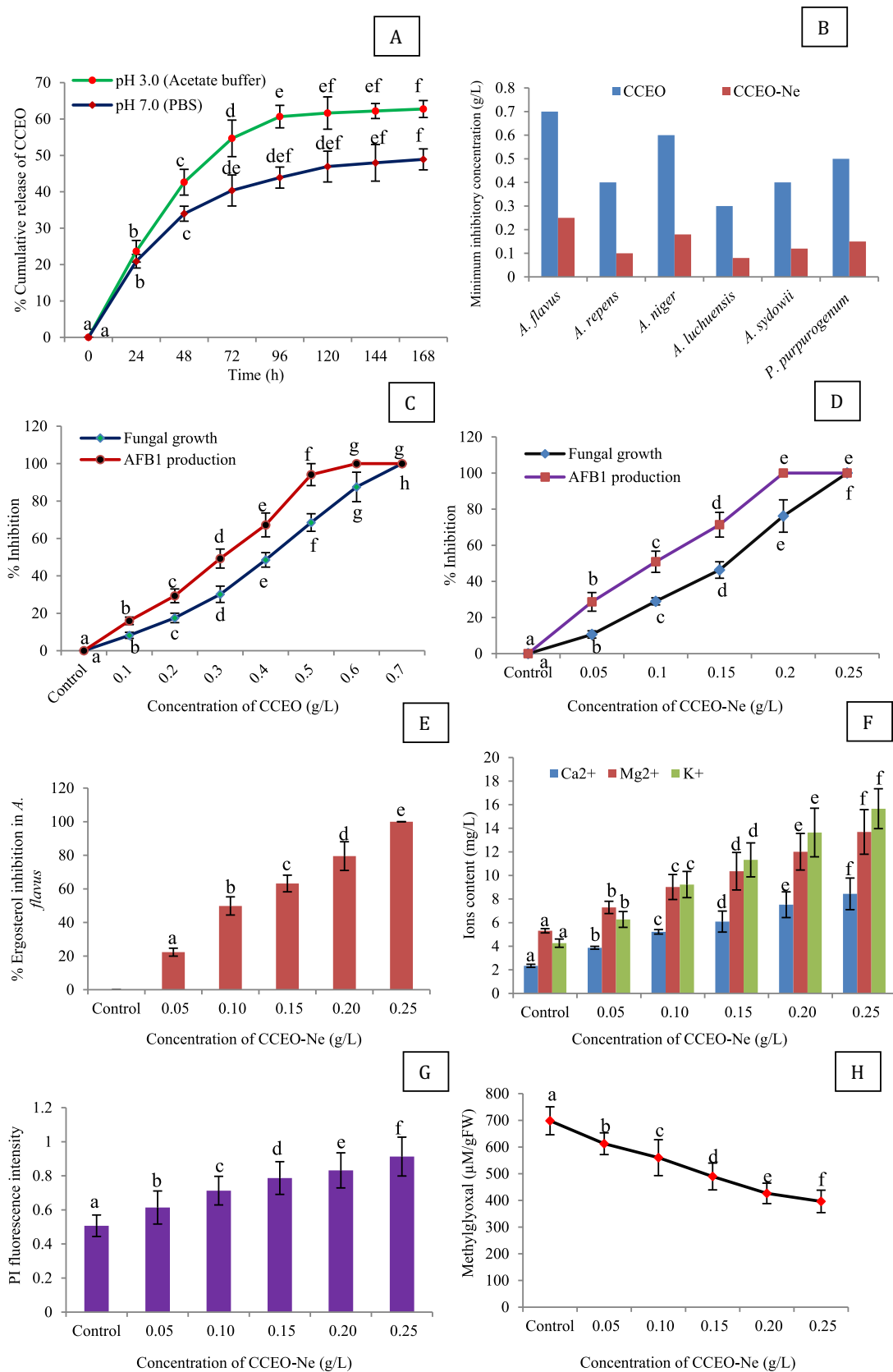


Fig. 2. (A) *In vitro* release of CCEO from chitosan matrix at acetate buffer and phosphate buffer saline, (B) minimum inhibitory concentration of CCEO and CCEO-Ne against broad range of fungi causing contamination of *Withania somnifera* root samples, (C) inhibition of *Aspergillus flavus* growth and AFB₁ production by CCEO, (D) inhibition of *Aspergillus flavus* growth and AFB₁ production by CCEO-Ne, (E) percent inhibition of ergosterol production in *A. flavus* cells by CCEO-Ne, (F) effect of CCEO-Ne on the leakage of ions from *A. flavus* cells, (G) effect of CCEO-Ne on membrane integrity of *A. flavus* cells by propidium iodide binding assay, (H) Effect of CCEO-Ne on methylglyoxal production in *A. flavus*.

3.2. Preparation of CCEO-Ne

The encapsulation of CCEO into chitosan nanomatrix was carried out by the synthesis of droplets followed by solidification. The CCEO droplet in chitosan solution was developed by homogenization based emulsion formation and reduction of surface energy by application of Tween-80 as stabilizing agent. The droplets were solidified by cross-linking of -NH_3^+ group of chitosan chain encompassing the CCEO droplets and PO_4^{3-} groups of S-TPP [43].

3.3. Encapsulation and loading efficiency of CCEO-Ne

The encapsulation and loading efficiency of CCEO-Ne were ranged from 56.37 to 95.52 % and 0.98–6.23 %, respectively (Table 2). On the basis of weight ratio of CCEO/chitosan, EE and LE showed upward trend (with addition of CCEO) until 1:0.8 ratio, while at 1:1 ratio the value was found to decrease significantly ($p < 0.05$). The decrement in EE and LE value was mainly attributed to saturation of CCEO entrapped into chitosan nanomatrix [44]. The finding of Hosseini et al. [45] obtained maximum LE of 2.12 % after entrapment of oregano essential oil into chitosan nanoparticle. Rajkumar et al. [46] also demonstrated highest EE of 40.62 % at 1:1 ratio of chitosan to *Piper nigrum* essential oil. The variation in the present finding with above mentioned results could be due to various factors like type of essential oil, polymer matrix, method of nanoemulsion synthesis, and relative proportions of different reagents [47]. Hence, the efficient imprisoning of CCEO offers a new insight in the synthesis of green fungitoxicant and maximizes their application as novel preservative of herbal raw materials in agriculture and pharmaceutical industries.

3.4. Characterization of Cs—Ne and CCEO-Ne

3.4.1. Zeta potential, size, and polydispersity index study

The zeta potential, size, and polydispersity index were measured through DLS. The particle size of Cs—Ne was 70.38 nm, whereas the size of CCEO-Ne was found to be 103.63 nm (Fig. 1A). This strongly suggests the incorporation of CCEO into chitosan particles. The result is in conformity with Taheri et al. [48] regarding entrapment of *Heracleum persicum* essential oil into chitosan nanoparticles. The zeta potential is considered as a prime factor for providing stability of the emulsion system which helps maintain particle repulsion in the emulsion system and prevent against coalescence and flocculation [49]. The zeta value near to ± 30 mV was suggested to be sufficient for stable emulsion system. The zeta potential of Cs—Ne was +37.95 mV suggesting positively charged particle surface. The zeta potential value of CCEO-Ne was decreased to +28.72 mV (Fig. 1B) reflecting the loading of CCEO and enhanced the dispersion stability of particles. This might be related with the shielding of charge by migration and coating of CCEO in nanoemulsion system. The PDI imparts the information about particle distribution in nanoemulsion system with homogeneity. The PDI of CCEO-Ne was found to be < 0.3 (Fig. 1C) validating the synthesis of mono-dispersed system with good homogeneity of distribution. Similar finding was obtained by Xavier et al. [50] indicating nano-entrapment of *Cinnamodendron dinisii* essential oil into zein matrix with homogeneity in particle distribution.

3.4.2. Morphological structures by SEM

Size distribution and morphology of Cs—Ne and CCEO-Ne was analyzed through SEM. The micro-structural details and surface characteristics were recorded at high magnifications. Spherical structure with smooth surface of Cs—Ne particles was recorded by SEM and particle sizes fall in the range of 48.51–76.81 nm (Fig. 1D). CCEO-Ne indicated increment in the size of particles (88.32–110.51 nm) (Fig. 1E) owing to loading of CCEO into chitosan nanobiopolymer. Similar report on size increment of clove essential oil incorporated to chitosan particles was described by Hasheminejad et al. [51]. The

Table 3

R^2 value for different kinetic models to show the release of CCEO.

Model names	R^2 (acetate buffer)	R^2 (phosphate buffer)
Zero order	0.776	0.790
First order	0.842	0.852
Higuchi model	0.941	0.961
Hixson	0.822	0.833
Korsmeyer-Peppas	0.967	0.964

average diameter of particles as measured by DLS was somewhat larger than SEM study because DLS measured the particles hydration radius which certainly was found larger as compared to SEM [52].

3.4.3. FTIR based characterization

FTIR was performed to evaluate the molecular interaction and cross-linking between functional groups of chitosan and CCEO along with S-TPP. The technique also verified the functional groups mediated characteristic vibrational modes after interaction with different components. The chitosan displayed characteristic absorption at 570, 658 (pyranoside ring stretching), 1056 (C-O-C bond stretching), 1380 (-OH group bending), 1590, 1660 (amide I C=O bond vibration), 2865 (stretching -CH bond), and 3440 cm^{-1} (-NH_2 and -OH stretching vibration) (Fig. 1F). The Cs—Ne showed similar peaks like chitosan, whereas shifting of amide I and amide II peaks were noticed at 1655 and 1586 cm^{-1} [53]. The peak of 1056 as presented in chitosan was slightly relocated to 1066 cm^{-1} . The additional characteristic peaks of Cs—Ne were observed at 1533 and 2916 cm^{-1} (Fig. 1G) owing to stretching of C—H bond at SP^3 carbon and tension of different C—O bonds. The spectra of CCEO was recorded at 730 (C—H bending), 840, 1017 (C-O-C stretching), 1446 (C=C bending), 1596 (C—H bond stretching), 1652, 1731 (C=O stretching), 2883, 2962 (C—H stretching), and 3425 cm^{-1} (-OH stretching) validating the presence of different components (Fig. 1H). Most notably, a close look to the CCEO-Ne emphasized the presence of most of the peaks of chitosan and CCEO with some stretching vibration and strong peak intensity (Fig. 1I) confirming that CCEO was successfully entrapped into chitosan nanomatrix [54].

3.4.4. XRD characterization

In XRD evaluation chitosan indicated two sharp peaks at 10.63 and 24.98° (Fig. 1J) representing high degree of crystallinity which shows similarity to Payandeh et al. [55]. A loss in crystallization degree of chitosan was recorded in case of Cs—Ne that might be due to trans-linking binding of chitosan to S-TPP [56]. The diffractogram peak of CCEO-Ne seemed to be narrower and destructed (Fig. 1J) with very less intensity indicating the incorporation of CCEO resulting in complex structural alteration of chitosan particles [24].

3.5. Release assay of CCEO

The release profile of CCEO was investigated in acetate buffer (pH 3.0) and phosphate buffer saline (pH 7.0) system (Fig. 2A). The release behavior is important because it confirms the successfulness of CCEO entrapment and demonstrates the mechanism of delivery. The result represented two phase release of CCEO viz. initial burst delivery superseded by controlled delivery. The explosion release of CCEO appeared during first 48 h in acetate and PBS media. The burst impact was contributed by diffusion of CCEO adhered at the surface of chitosan particles, high rate of dissolution of chitosan matrix near the surface, and quick diffusion of CCEO upon contact with discharged environment [57]. However, CCEO showed dramatic decrement in delivery after 48 h and reached a plateau after 96 h. The controlled release of CCEO at subsequent phases may be explained by concentration gradient of CCEO between buffer and chitosan particles together with inability of buffer in breaking the compact chitosan structures [58]. It has been pointed that acetate buffer showed greater release of CCEO over PBS which may be

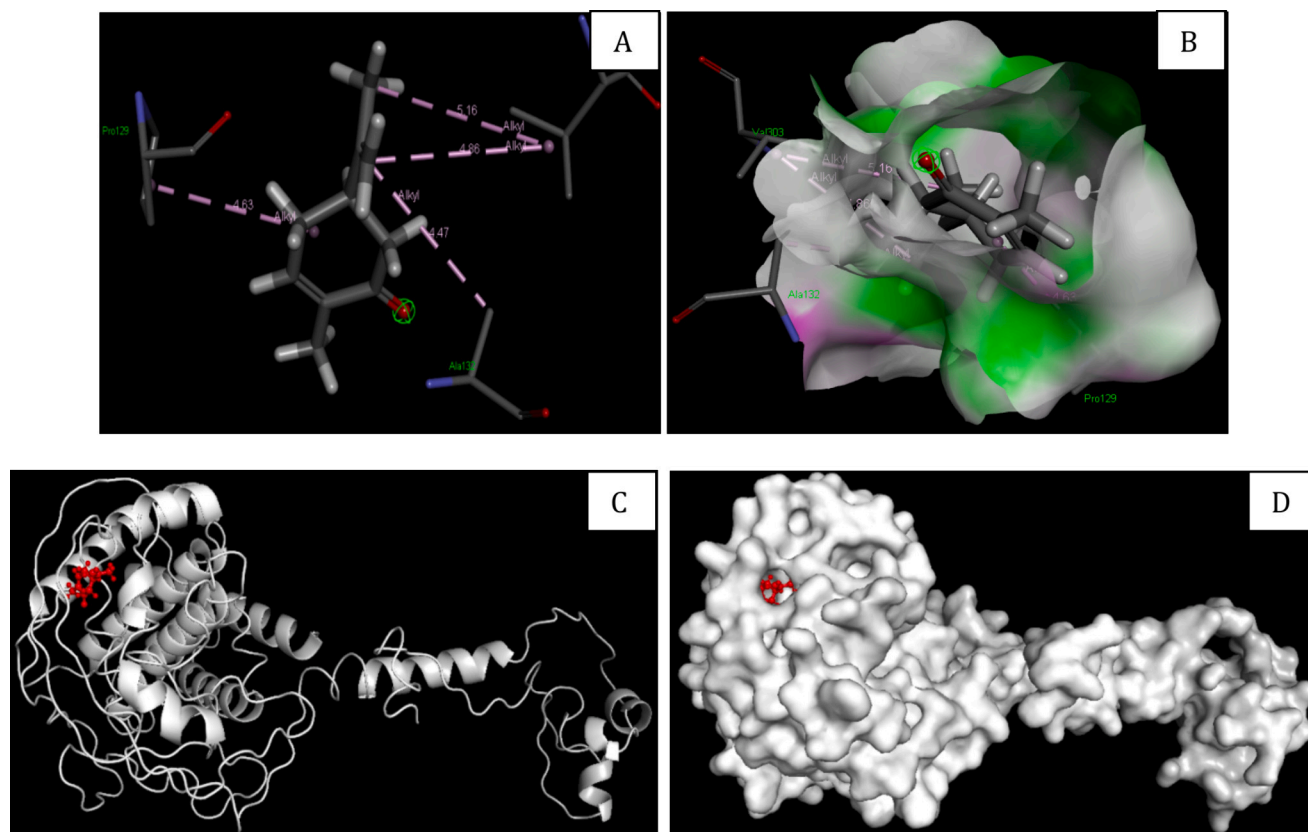


Fig. 3. (A, B) Binding of carvone with Afl-R protein, (C, D) Internal pocketing of carvone to the active site of Afl-R protein.

attributed to the greater dissolution and swelling of hydrophilic nanoparticles in acidic medium [24]. Moreover, ionic repulsion of free amino group in one chitosan chain with another chain and subsequent diffusion of entrapped oil may be a possible reason for sustained delivery of CCEO. The release process is closely matched with Korsmeyer-Peppas model for acetate and phosphate buffer saline having R^2 values equivalent to 0.967 and 0.964, respectively. Other kinetic models demonstrated lower value of R^2 in comparison to Korsmeyer-Peppas (Table 3). The model suggests non-fickian process of release of essential oils entrapped in polymeric matrix comprising of chitosan [59]. Further, the Korsmeyer-Peppas model can be applied successfully not only in the cases involving release of encapsulated compounds through more than mechanisms, but also in conditions of unknown release kinetics [60,61]. Similar biphasic pattern of fast release followed by controlled delivery of clove essential oil from chitosan nanoparticle has also been presented by Hasheminejad et al. [51]. The present result demonstrates the controlled release of CCEO which could maintain the gradual antifungal effect in stored herbal raw materials.

3.6. In vitro evaluation of antifungal and anti-AFB₁ activity

The MIC of CCEO against *A. flavus*, *A. sydowii*, *A. luchuensis*, *A. repens*, *A. niger*, and *Penicillium purpurogenum* was recorded to be 0.7, 0.4, 0.6, 0.3, 0.4, and 0.5 g/L, respectively (Fig. 2 B). The antifungal activity of CCEO was found better than previously presented *Cymbopogon flexuosus* essential oil which inhibited the growth of varieties of fungi contaminating different herbal raw materials [62]. In case of synthetic preservatives salicylic acid inhibited the fungal growth at 3.0 g/L, while butylated hydroxyl toluene (BHT), gallic acid, and ascorbic acid showed very poor antifungal activity at 10 g/L [63]. The Cs—Ne displayed negligible inhibition of different fungal species as compared to free and nanoencapsulated CCEO. This effect of chitosan has been connected to the fact that chitosan does not hold ability to diffuse in the media, hence,

fungal cells which are in direct contact with the chitosan's active site are inhibited [64]. Moreover, the performance of antifungal activity of chitosan requires the interaction of positively charged amino group of chitosan with different anionic groups of fungal cell surface inhibiting the synthesis of mRNA and protein via penetration into cells and formation of a barrier, thereby suppressing the absorption of fungal nutrients [65]. During the encapsulation process most of the free amino groups of chitosan are utilized after interaction with polyphosphate groups of S-TTP and thus are unavailable to react with fungal cell surface which might be a possible reason for negligible antifungal activity of Cs—Ne. The CCEO-Ne displayed superior inhibition of *A. flavus*, *A. repens*, *A. niger*, *Penicillium purpurogenum*, *A. sydowii*, and *A. luchuensis* at 0.25, 0.1, 0.18, 0.08, 0.12, and 0.15 g/L, respectively (Fig. 2 B). The CCEO-Ne exhibited nearly 3–4 times better performance in the inhibition of different fungal growth as compared to unencapsulated CCEO. The nanometric size of CCEO-Ne particles with physical stability and controlled delivery mainly contributed to the improvement in antifungal efficacy [66]. The better growth inhibition of different fungi validated enhancement in bioactivity over free essential oil. The MAIC of CCEO against AFB₁ was 0.6 g/L (Fig. 2C). Salicylic acid caused complete inhibition of AFB₁ production at 3.0 g/L, while the ascorbic acid and gallic acid at 10 g/L. The BHT was found as a poor AFB₁ inhibitory agent causing 61.68 % inhibition at 10 g/L [63]. In a recent investigation of Bhavitha et al. [67] ozone based standard practices for inhibition of AFB₁ has also been demonstrated. The result indicated that ozone supplementation at different concentrations (0.02–0.13 g/L) delayed and decreased the AFB₁ production, however, the treatment could not suppress AFB₁ production completely. Complete suppression of AFB₁ production by CCEO-Ne was found at 0.2 g/L (Fig. 2D). The superiority in inhibition of AFB₁ production over fungal growth suppression might be connected with impaired catabolism of carbohydrate by norsolorinic acid dependent key enzymes in *A. flavus* leading to suppression of AFB₁ synthesis [68]. Most importantly, the encapsulation protected the CCEO

Table 4
Binding interaction of carvone with Afl-R protein.

Component	Receptor protein	Hydrogen bonding amino acids	Binding energy indices (kcal/mol)			
			Global energy	Attractive Van der Waal's force	Repulsive Van der Wall's force	Atomic contact energy
Carvone	Afl-R	Pro 129, Ala 132, and Val 303	−30.21	−21.32	4.87	−11.68

Table 5
Antioxidant activity of CCEO, Cs—Ne, and CCEO-Ne.

Essential oil	DPPH IC ₅₀ value (g/L)	ABTS IC ₅₀ value (g/L)
CCEO	10.56 ± 1.03 ^a	8.69 ± 0.96 ^a
Cs-Ne	45.97 ± 3.87 ^b	37.96 ± 3.95 ^b
CCEO-Ne	7.01 ± 0.84 ^c	4.23 ± 0.92 ^c

Note: Values are mean (n = 3) ± SE. The means followed by same letter in the same column are not significantly different according to ANOVA and Tukey's multiple comparison tests.
Absorbance of negative control for both assays was determined as 0 % inhibition.

from external influence of environmental factors and has been suggested as an innovative strategy to enhance the efficacy of CCEO.

3.7. Antifungal mechanism of action of CCEO-Ne against *A. flavus*

Here, *A. flavus* was recognized as fast spreading and AFB₁ producing

fungi causing contamination of *Withania somnifera* roots, therefore, antifungal mechanism of CCEO-Ne against *A. flavus* is of prime importance for the current study. The Cs—Ne did not show any prominent antifungal performance; hence CCEO-Ne was taken into consideration for elucidating the target site of action in *A. flavus* cells. Target site of CCEO-Ne against *A. flavus* was determined through ergosterol content, ions leakage, and PI binding assay. Ergosterol is a principal sterol compound in fungal plasma membrane which stabilizes the membrane by phospholipid binding and maintains the membrane fluidity and permeability along with regulation of substance transportation and membrane dependent enzymatic activities [69,70]. Ergosterol content was found to decrease in a dose dependent fashion after CCEO-Ne treatment. The ergosterol content was reduced by 22.34, 49.85, 63.21, 79.52, and 100 % at 0.05, 0.10, 0.15, 0.20, and 0.25 g/L of CCEO-Ne, respectively (Fig. 2E). The suppression of ergosterol synthesis has been attributed to inhibition of 14- α demethylase activity during lanosterol mediated ergosterol formation [71]. Moreover, the encapsulation of CCEO in the form of nanoemulsion (nano-sized) supported our hypothesis of greater efficacy in ergosterol inhibition. The CCEO-Ne also

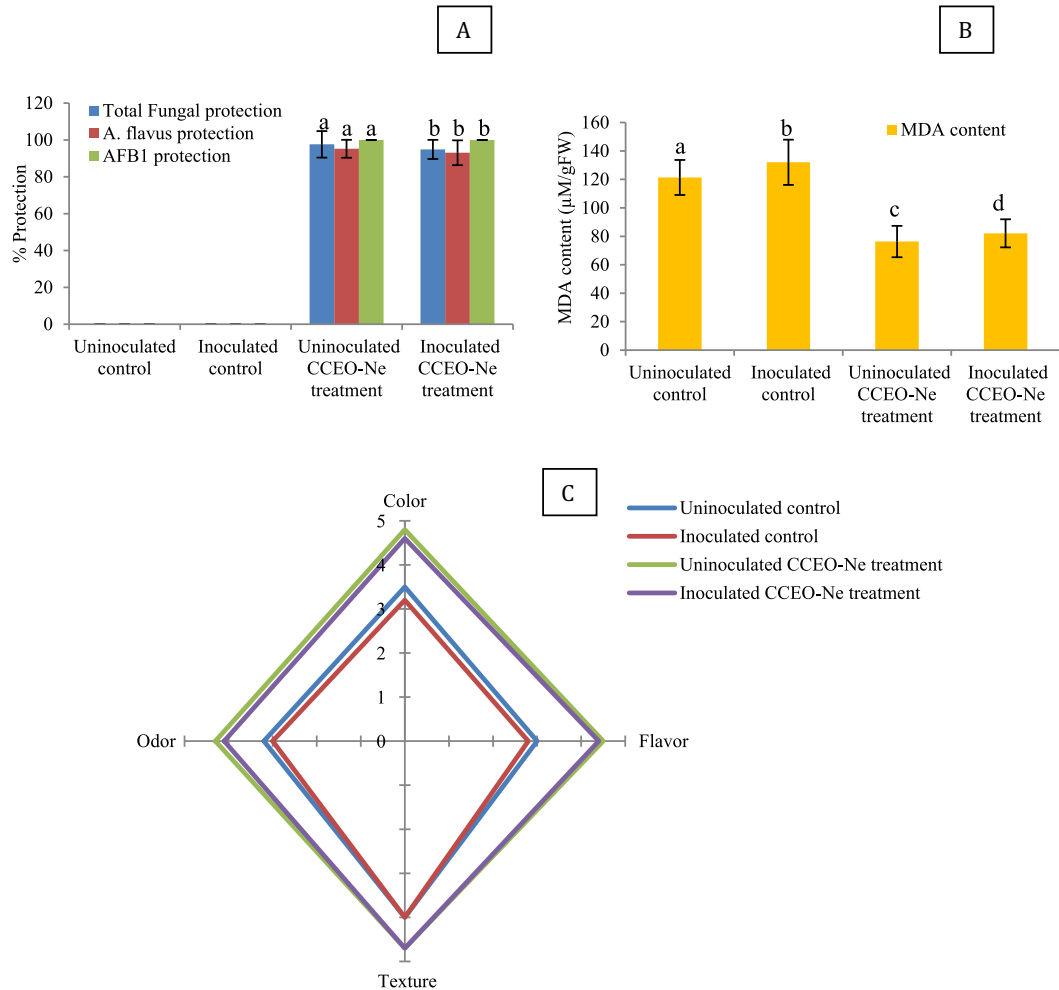


Fig. 4. (A) *In situ* antifungal and anti-aflatoxigenic protection of *W. somnifera* root samples by CCEO-Ne, (B) Effect of CCEO-Ne on MDA content of *W. somnifera* root samples, (C) Effect of CCEO-Ne on organoleptic properties of *W. somnifera* root samples.

caused leakage of K^+ , Ca^{2+} , and Mg^{2+} ions at different concentrations (Fig. 2F) confirming alteration in plasma membrane permeability by targeted action. Most notably, the membrane integrity in *A. flavus* cells was validated by PI assay. PI is a fluorescing stain which could pass into injured cells and binds with DNA, while the healthy cell inhibits the entrance of PI due to membrane barrier [72,73]. The Fig. 2G shows very less PI fluorescent intensity at control set, while at different concentrations of CCEO-Ne fluorescent intensity was increased significantly ($p < 0.05$). Enhancement in PI fluorescence suggested the irreversible destruction of plasma membrane integrity. Hence, the antifungal action of CCEO-Ne was associated with disruption of membrane permeability and integrity. The damage of membrane could lead the impairment in cellular respiration, blockage of enzyme mediated reactions, and cell death. However, the interaction of CCEO nanoemulsion with plasma membrane receptors and channel proteins would be done in future research for better elucidation of antifungal mechanism of action.

3.8. Biochemical and molecular mechanism of AFB₁ suppression by CCEO-Ne

Biochemical mechanism of AFB₁ suppression was determined by methylglyoxal assessment. Methylglyoxal is a reactive carbonyl species formed by lipids, proteins, or carbohydrates degradation and highly toxic to nucleic acid and proteins (especially reacts with arginine and lysine) due to formation of advanced glycation end products [74]. Methylglyoxal is the current hotspot of cell biology because of its potential role to induce AFB₁ biosynthesis by upregulating the activity of *afl-R*, *nor-1*, and *omt-A* genes, and serving as a master link between AFB₁ production and cellular metabolism [75]. The CCEO-Ne displayed concentration dependent reduction of methylglyoxal with the content 396.25 μ M/gFW at 0.25 g/L as compared to control having methylglyoxal level 698.32 μ M/gFW (Fig. 2H). The carvone dependent methylglyoxal trapping may be a possible reason for reduced synthesis of methylglyoxal [76]. The result demonstrated the role of CCEO-Ne as novel methylglyoxal suppressor which in turn reduced the level of AFB₁ in *A. flavus* cells. Hence, the decrement in methylglyoxal level could be considered for developing innovative preservation strategies in pharmaceutical industries by sustainable transgenic technology.

Investigation on molecular mechanism of anti-AFB₁ activity was performed through *in silico* molecular docking of Afl-R protein with carvone. The Afl-R is a zinc cluster (Zn(II)₂Cys₆) protein which has the ability to bind specific DNA sequence and helps in structural genes transcription [77]. Disruption of this enzymatic activity results in the inhibition of the AFB₁ production. The modeling validated the carvone dependent interaction of Afl-R protein by hydrogen bonding of different amino acids like Pro 129, Ala 132, and Val 303 (Fig. 3A, B). The fig. C and D also demonstrated the internal pocket binding of carvone with Afl-R protein. The interaction suitability was confirmed through binding energy parameters viz. global energy, atomic contact energy, repulsive and attractive Van der Waal's force (Table 4). More negative binding energy suggested the H-bond dependent stereo-spatial interaction of carvone at the active site of Afl-R protein. The present result is corresponded to the work of Murugan et al. [78] determining interaction of methyl palmitate, 3-hexane 1-yl benzoate, and 3,5 dimethylcyclohexanol with Ver-1 protein responsible for AFB₁ reduction. The *in silico* result could provide a green insight for the development of essential oil based fungitoxicant possessing better agricultural and pharmaceutical application. However, the wet lab experiments on application of CCEO nanoemulsion against the genes governing the AFB₁ secretion like *afl R*, *ver 1*, and *omt A* genes by real time polymerase chain reaction (RT-PCR) would be conducted in future studies for better evaluation of anti-aflatoxigenic mechanism of action.

3.9. Evaluation of antioxidant activity

Fungal contamination leads to production of reactive oxygen species

(ROS) causing biodeterioration of active constituents like withanolide and withaferin A in herbal raw materials. The ROS are involved in inducing the AFB₁ generation in stored herbal raw materials by free radical mediated chain reactions [4]. Therefore, the free radical quenching potentiality of CCEO-Ne is crucial to be determined before practical application. The antioxidant potential of CCEO-Ne was evaluated through DPPH and ABTS quenching process. The 50 % quenching potentiality of CCEO-Ne was recognized as 7.01 and 4.23 g/L by DPPH and ABTS assay, respectively (Table 5). The CCEO-Ne showed greater antioxidant activity over the unencapsulated CCEO (DPPH IC₅₀ = 10.56 g/L and ABTS IC₅₀ = 8.69 g/L). The IC₅₀ value of Cs—Ne was considerably high against DPPH (45.97 g/L) and ABTS (37.96 g/L) assay, suggesting less antioxidant potentiality. Better effectiveness in antioxidant quality has been linked with nano-size of CCEO droplets immersed in chitosan nanoemulsion with greater surface area. Moreover, the sustained delivery of essential oil from chitosan matrix led to enhanced exposure towards the pool of free radicals with greater bioavailability resulting in the enhancement of antioxidant activity. Woranuch and Yoksan [79] demonstrated escalation in free radical neutralizing potentiality of eugenol after imprisoning into chitosan nanoparticles which has been linked with protection against adverse effects like temperature and oxygen. The outcome of current study is corresponded to the finding of Hadidi et al. [80] showing improvement in free radical neutralizing capability of clove oil after encapsulation into chitosan polymer. On the basis of this result, the CCEO-Ne may serve as an alternative of synthetic antioxidants to preserve the herbal raw materials against AFB₁ contamination.

3.10. In situ effectiveness evaluation of CCEO-Ne in stored herbal raw material (*Withania somnifera* roots)

The *in situ* antifungal and AFB₁ suppressing effectiveness of CCEO-Ne was performed in *Withania somnifera* roots as a model herbal raw material to assess the practical efficacy after 1 year of storage period. The CCEO-Ne caused 97.63 and 94.87 % protection of *W. somnifera* roots against fungal contamination in uninoculated and inoculated treatment, respectively; while the protection against *A. flavus* was recorded as 95.22 and 93.06 % in uninoculated and inoculated treatment sets, respectively (Fig. 4A). The 100 % protection of root samples against fungal inhabitation was not achieved which may be linked with absorption of some CCEO components by herbal samples. Two control and treatments were designed to assess the inoculum density and potential regarding production of AFB₁ in stored *W. somnifera* roots. The AFB₁ content in *W. somnifera* roots were measured through HPLC assay and AFB₁ level was found as 15.23 and 12.64 μ g/kg in inoculated and uninoculated root samples. The inoculated and uninoculated treatment (CCEO-Ne fumigated) showed 100 % reduction of AFB₁ in root samples (Fig. 4A). The disparity in fungal association and AFB₁ synthesis has been associated with alteration in carbohydrate catabolism, a prime step in AFB₁ synthesis. Moreover, the reduction of methylglyoxal content in *A. flavus* also play prominent role in the better alleviation of AFB₁ production in *W. somnifera* roots. Superior *in situ* performance of CCEO-Ne has been associated with shielding of essential oil against oxidation and sustained release pattern with increasing surface area/volume ratio, dispersibility, and stability [81]. The present outcome is coincided with findings of Singh et al. [26] demonstrating the protection of *Picrorrhiza kurroa* roots deterioration by nanoencapsulated *Ocimum sanctum* essential oil.

3.11. Effect of CCEO-Ne on lipid peroxidation of *W. somnifera* roots

Lipid peroxidation is a prime factor for deterioration of active constituents of herbal raw materials. Peroxidation has been linked with generation of ROS which in turn reacts with polyunsaturated fatty acids [82]. The AFB₁ contamination in herbal raw materials has also been responsible for lipid peroxidation affecting the organoleptic properties

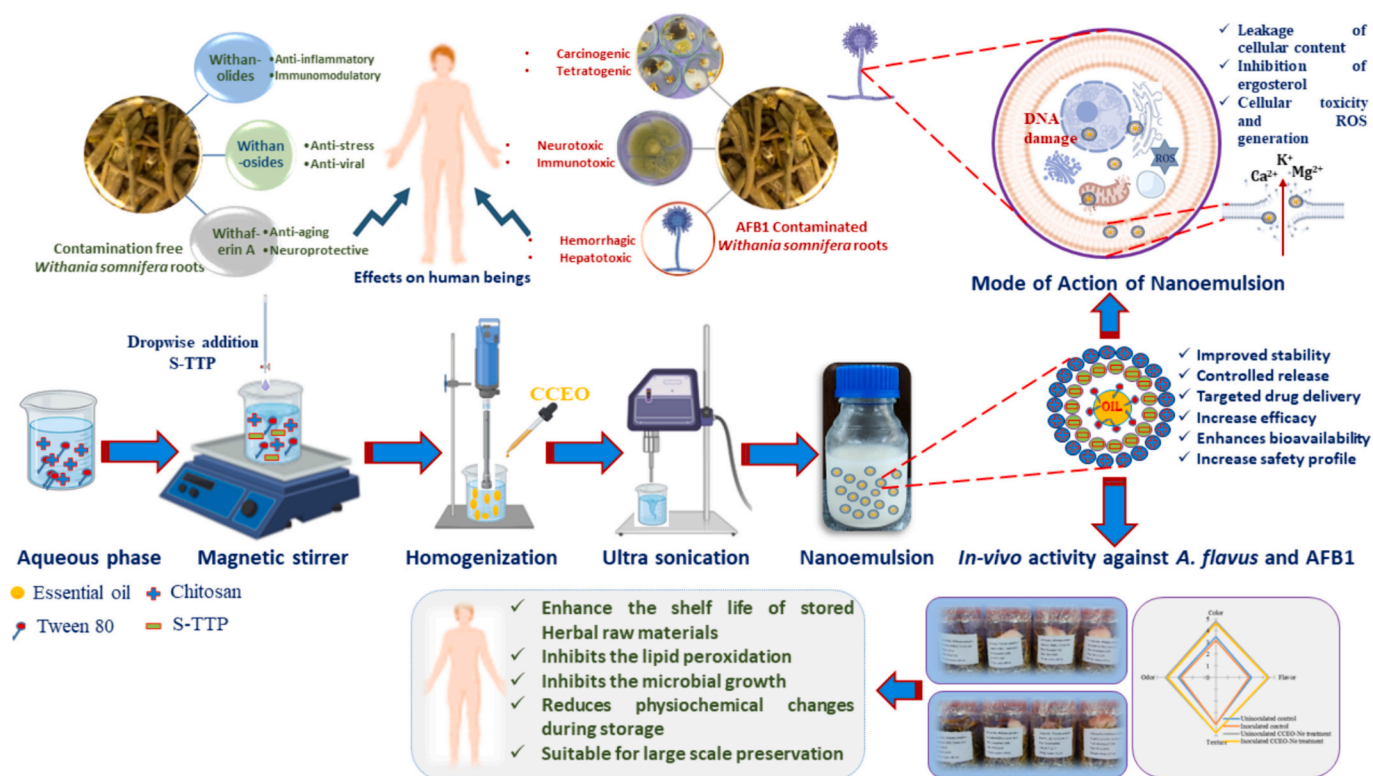


Fig. 5. Scheme based outline of *Withania somnifera* roots contamination by *A. flavus* and AFB₁ biosynthesis together with mitigation of roots deterioration by nanoencapsulated *Carum carvi* essential oil and green insight to quality maintenance. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

[4]. Moreover, the lipid peroxidation depends on fatty acid composition, pH, concentration of oxygen, and distribution of ions. MDA content in inoculated and uninoculated *W. somnifera* root samples were displayed as 132.03 and 121.36 µg/kg, respectively; however, CCEO-Ne showed prominent reduction in MDA content to 82.10 and 76.32 µg/kg for inoculated and uninoculated CCEO-Ne treatment sets at MIC concentration, respectively (Fig. 4B). CCEO-Ne dependent suppression of lipid peroxidation was mainly attributed to controlled delivery and better free radical scavenging potentiality. The result is similar to the investigation of Singh et al. [83] regarding effective reduction of lipid peroxidation by nanoencapsulated *Ocimum americanum* essential oil in stored millet samples after 1 year of storage. Hence, the current finding strengthens CCEO-Ne application for protection of herbal raw materials.

3.12. Organoleptic property assessment

Organoleptic property evaluation plays an important role in the acceptability of herbal raw materials for consumers. During the course of investigation, significantly lower scores were recorded for color, flavor, and odor of uninoculated and inoculated control *W. somnifera* root samples (Fig. 4C). Fungal and AFB₁ contamination did not display any significant effect on texture of root samples. Deterioration in flavor, color, and odor of control root samples was mainly happened due to infestation of fungi and AFB₁ production in herbal raw materials. The AFB₁ may also induce the generation of ROS which could react with active constituents of *W. somnifera* roots and develop off-flavor, off-odor, and bitter taste. However, the CCEO-Ne fumigation maintained the color, flavor, odor, and texture of root samples over 1 year of storage period (Fig. 4C). Preservation of sensory properties of root samples by CCEO-Ne has been linked with better quenching potentiality of free radicals and inhibition of methylglyoxal production which in turn facilitated the suppression of AFB₁ synthesis in due course of time. In a study, Chaudhari et al. [84] exemplified the preservation of sensory

properties of maize samples treated with nanoencapsulated *Pimenta dioca* essential oil. Our current finding demonstrated the fact that nanoencapsulation of CCEO could facilitate in reduction of AFB₁ mediated herbal materials degradation without altering the sensory properties which promptly signifies its industrial relevance.

3.13. Evaluation of safety profile in animal model

To ensure the application of CCEO-Ne as preservative in agricultural and pharmaceutical industries, safety evaluation was performed in albino mice. The mortality of mice was recorded after 48 h at different concentrations of CCEO-Ne (50–600 µL/31.63 g body weight). The 50 % mortality value (LD₅₀) was recorded from dose response logarithmic regression curve and converted into lethal dose through probit analysis. LD₅₀ of CCEO-Ne was calculated as 12,089.23 µL/kg body weight. Lethal dose was greater than predominantly used synthetic preservatives like tetraethylpyrophosphate, parathion, and fonophos (0.01–0.025 g/kg) and botanical fumigants viz. pyrethrum (300–500 mg/kg) and azadirachtin (5000 mg/kg) [85]. Most importantly, the LD₅₀ value exceeds the OECD cut-off level (2 g/kg) which strengthen its utilization as non-toxic green chemical. Hence, the nanoemulsion can be recommended as safe botanical preservative with new insight to maximize their application for preserving herbal raw materials. A scheme based outline of *Withania somnifera* roots contamination by *A. flavus* and AFB₁ biosynthesis together with mitigation of roots deterioration by encapsulated *Carum carvi* essential oil nanoemulsion and green insight to quality maintenance is displayed in Fig. 5.

4. Conclusion

The nanoencapsulation led to improvement in antifungal activity of CCEO, thereby preventing deterioration of *W. somnifera* root samples by fungi and AFB₁ contamination. The mechanism of inhibition of toxigenic

A. flavus was attributed to impairment in ergosterol synthesis, leakage of cellular ions, and destruction of membrane integrity. Inhibition of methylglyoxal production along with modeling based *in silico* study validated underlying mechanism for suppression of AFB₁ biosynthesis. Further, the *in situ* antifungal and anti-AFB₁ protection of *W. somnifera* root samples together with inhibition of lipid peroxidation without alteration of organoleptic qualities and safety profile in mice model reinforced the application of nanoencapsulated CCEO emulsion as green shelf-life enhancer to preserve the herbal raw materials. Considering the industrial relevance, it is suggested that future studies should focus more on optimization of encapsulation methodology to enhance the encapsulation efficiency and loading capacity, hence applying the minimum concentration of essential oil for practical application. In addition, detailed investigations are required to understand the molecular pathways and specific genes governing the AFB₁ biosynthesis for essential oil nanoformulation based alleviation of AFB₁ contamination in herbal raw materials. Further, more research is warranted for determination of essential oil based nanomaterial toxicity, effect on non-target organism, human health, and regulatory framework to improve the market status and better commercialization in agricultural and pharmaceutical industries.

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CRediT authorship contribution statement

Somenath Das: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Akash Maurya:** Validation, Software, Resources, Investigation, Formal analysis. **Vipin Kumar Singh:** Writing – review & editing, Visualization, Validation. **Anand Kumar Chaudhari:** Methodology, Investigation. **Bijendra Kumar Singh:** Formal analysis, Data curation. **Abhishek Kumar Dwivedy:** Investigation, Formal analysis. **Nawal Kishore Dubey:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare no conflicts of financial and competing interest.

Data availability

The data that has been used is confidential.

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